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Industrial Systems Modernization | Manufacturing Data & Automation

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Candidate Thesis Brief

Industrial Automation Engineer (Ignition Development) · Bytecubit Technologies

EXECUTIVE THESIS

Don't migrate screens. Prove the signal path.

Run legacy and target behavior side by side, reconcile data context, and cut over only when function, data, ownership, and rollback evidence are explicit.

ACTUAL MANDATE

Modernize legacy MES functionality with Ignition while protecting production continuity; improve SQL-backed manufacturing applications; standardize the AVEVA PI historian environment; and create dependable manufacturing data integration into a cloud data lake.

WHY THIS MATTERS NOW

Legacy functionality

Business-critical behavior may be encoded across screens, queries, scripts, workarounds, and local knowledge.

Historian entropy

Duplicate, stale, poorly named, or weakly contextualized signals increase support burden and reduce analytic trust.

Cloud demand

Raw plant signals are not automatically usable data products; context, ownership, quality, and semantics must travel with them.

Contract clock

The team needs visible progress through year-end without sacrificing maintainability for speed.

CENTRAL CONTRADICTIONS

- Fast delivery vs. production continuity.
- Reusable Ignition standards vs. legitimate local process differences.
- Historian cleanup vs. unknown downstream dependencies.
- Cloud accessibility vs. OT reliability, ownership, and security.

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Signal Twin Workbench

A cutover method for MES modernization, historian cleanup, and plant-to-cloud data integrity.

THE MECHANISM

Stage	Question	Evidence	Decision
Read	What does the live function actually do?	States, exceptions, users, SQL, tags, consumers, failure behavior	Bound the scope
Twin	What is the smallest supportable Ignition replacement?	UDTs/tags, components, logic, interfaces, diagnostics, source control	Build / revise
Reconcile	Does target data carry stable meaning?	Naming, timestamps, units, quality, PI context, SQL schema, ownership	Accept / repair
Prove	Does the target behave under real conditions?	Parity tests, exceptions, performance, operator acceptance, rollback	Promote / hold
Release	Can the team support and learn from it?	Runbook, owner, observation, incident path, reusable pattern	Cut over / roll back

ILLUSTRATIVE SCENARIO STATES

Legacy function Focus: hidden behavior and consumers. Minimum evidence: state/event/context/rollback parity.	Historian cleanup Focus: point usage, stewardship, AF context, retention, and consumer impact before deletion or renaming.
New line onboarding Focus: reusable tag/UDT/application templates without scaling an unproven local pattern.	Cloud analytics feed Focus: freshness, semantics, quality, access, and named data-product ownership.

RELEASE CRITERIA

Function parity · Data integrity · Operator usability · Performance · Security/access · Support ownership · Rollback readiness · Post-release observation.

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Role Alignment Map

Verified evidence selected for the actual mandate—not keyword volume.

EVIDENCE-TO-MANDATE ALIGNMENT

Mandate need	Verified evidence	Transfer logic
Manufacturing environment	Vape-Jet software-enabled robotics; Compunetics advanced electronics; Amazon continuous operations	Understands frontline consequence, production continuity, quality, escalation, and live-system change.
MES/application modernization	ERP/Odoo and cross-workflow transformation; remote commercial/fulfillment modernization; standard-work and process-control mechanisms	Can map current work, define target state, create ownership, and install repeatable mechanisms.
SQL/data workflows	SQL in technical toolkit; data, telemetry, dashboards, workflow and agentic-system design	Can connect application behavior to schemas, operational context, evidence, and downstream decisions.
Industrial reliability	High-reliability electronics, quality systems, root cause, safety record, 24/7 operations	Brings consequence-aware testing, cutover, rollback, and support discipline.
AI/agentic potential	Agentic workflows, RAG, human authority, evaluation, governance, Python	Can add AI only where context, risk, human authority, and measurable work justify it.

EXPERIENCE BOUNDARY

Platform-specific boundary: my verified record does not include years of production Ignition configuration, direct AVEVA PI administration tenure, or ownership of a named MES cutover. The alignment rests on verified industrial systems, manufacturing, SQL/data, modernization, reliability, and cross-functional execution experience.

CANDIDATE ADVANTAGE

I can translate across manufacturing, engineering, quality, operations, databases, support, leadership, and user adoption. That breadth matters when the failure mode is not coding alone but a disconnected application that operators cannot trust, support cannot diagnose, or analytics cannot interpret.

[Open the full role alignment brief →](#)

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Operating Alignment

Priorities, questions, and success conditions for the work.

FIRST CONTRIBUTION SEQUENCE

1. Select a representative legacy function or bounded SQL/tag path.
2. Map the live states, dependencies, consumers, ownership, and failure modes.
3. Design the target Ignition component, tag/UDT structure, database interaction, and diagnostics.
4. Define parity, performance, exception, usability, rollback, and observation criteria.
5. Capture the reusable pattern and identify what must remain plant-specific.

QUESTIONS I WOULD BRING INTO THE ROOM

- Which functions must be modernized by year-end, and why those first?
- What is the current Ignition architecture and deployment discipline?
- Where is ownership ambiguous across manufacturing, controls, IT, data, R&D, and vendors?
- Which PI cleanup decisions are reversible, and which require a consumer-impact map?
- What does the cloud data lake need to enable that current plant systems cannot?
- What would make a contract extension an obvious business decision?

SUCCESS MEASURES

Validated function releases; production disruption avoided; recurring support issues reduced; cleaner data context; faster diagnosis; reusable templates; accountable owners; accepted cloud data products; learning captured for the next plant or function.

CLOSING POSITION

The role should leave more than working code. It should leave a more legible manufacturing system: explicit behavior, trustworthy data, known ownership, safer releases, and a repeatable modernization method.